# Arjuna JEE 2027

## Physics by Rajwant Singh Sir

## Units and Measurements

DPP: 11

## Time duration - 30 min

Q1 The most accurate reading of the length of $6.28\ cm$ long fibre is
(A) $6\ cm$
(B) $6.5\ cm$
(C) $5.99\ cm$
(D) $6.0\ cm$

Q2 The number of significant figures in $0.06900$ is:
(A) 5 (B) 4
(C) 2 (D) 3

Q3 The number of significant figures in $0.0006032\ m^2$ is:
(A) 4 (B) 5
(C) 7 (D) 3

Q4 What is the number of significant figures in $0.310 \times 10^3$ ?
(A) 2 (B) 3
(C) 4 (D) 3

Q5 The sum of the numbers $436.32, 227.2$ and $0.301$ in appropriate significant figures is:
(A) 663.821 (B) 664
(C) 663.8 (D) 663.82

Q6 Taking into account the significant figures, what is the value of $(9.99\ m - 0.0099\ m)$ ?
(A) $9.98\ m$
(B) $9.980\ m$
(C) $9.9\ m$
(D) $9.9801\ m$

Q7 The decimal equivalent of $1/20$ upto three significant figures is
(A) 0.0500
(B) 0.05000
(C) 0.0050
(D) $5.0 \times 10$

Q8 The numbers $2.745$ and $2.735$ on rounding off to 3 significant figures will give:
(A) $2.75$ and $2.74$
(B) $2.74$ and $2.73$
(C) $2.75$ and $2.73$
(D) $2.74$ and $2.74$

Q9 The length, breadth, and thickness of a block are given by $l = 12\ cm$, $b = 6\ cm$ and $t = 2.45\ cm$. The volume of the block according to the idea of significant figures should be:
(A) $1 \times 10\ cm^3$
(B) $2 \times 10^2\ cm^3$
(C) $1.764 \times 10\ cm^3$
(D) None of these

Q10 The mass and volume of a body are $4.237\ g$ and $2.5\ cm^3$, respectively. The density of the material of the body in correct significant figures is:
(A) $1.6048\ g\ cm^{-3}$
(B) $1.69\ g\ cm^{-3}$
(C) $1.7\ g\ cm^{-3}$
(D) $1.695\ g\ cm^{-3}$

Q11 Subtract $2.5 \times 10^6$ from $4.0 \times 10^{-4}$ with due regard to significant figures
(A) $4.0 \times 10^{-4}$
(B) $6.0 \times 10^4$
(C) $4.0 \times 10^{-3}$
(D) $4.0 \times 10^5$

Q12 The mass and radius of the earth are $5.975 \times 10^{24}\ kg$ and $6.37 \times 10^6\ m$

(A) $5.53 \times 10^3$
(B) $5.52 \times 10^3$
(C) $5.54 \times 10^3$
(D) $5.5 \times 10^3$

respectively. Calculate the average earth's density to correct significant figures. Take pi = 3.142

(A) $5.53 \times 10^{3}$

(B) $5.52 \times 10^{3}$

(C) $5.54 \times 10^{3}$

(D) $5.5 \times 10^{3}$

Q13 The significant figures of the number $6.0023$ are:

(A) 1 (B) 5

(C) 4 (D) 2

Q14 If $L\ 2.331\ cm = B\ 2.1\ cm$ = then $L + B$ =

(A) 4.423 cm

(B) 4.43 cm

(C) 4.4 cm

(D) 4 cm

## Answer Key

Q1 (B)

Q2 (B)

Q3 (A)

Q4 (B)

Q5 (C)

Q6 (A)

Q7 (A)

Q8 (D)

Q9 (B)

Q10(C)

Q11(A)

Q12(B)

Q13(B)

Q14(C)